

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q24-2399

Palo Duro 230 kV
September 03, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2399. The Interconnection Customer proposes a 24.0 MW natural gas combined cycle facility interconnecting at the Palo Duro 230 kV substation in the Ohio area. The request was received on April 10, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 422 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$150,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Hartland 230 kV/1 kV Auto
Contingency: Hubbardstown-Geyer Switch 345.0 kV Line
Loading: 101.1% (Rating: 270 MVA)
DFAX: 10.9%
Pre-project loading: 100.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hartland 230 kV/1 kV Autotransformer	Replace transformer with higher-rated unit	\$11,487,000	1% (\$150,000)
	Total allocated to project		\$150,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.